

Problem 9 [MEDQA]: Question

Question:

A 23-year-old woman comes to the physician because she is embarrassed about the appearance of her nails. She has no history of serious illness and takes no medications. She appears well. A photograph of the nails is shown. Which of the following additional findings is most likely in this patient?

Options:

- A. Silvery plaques on extensor surfaces
- B. Flesh-colored papules in the lumbosacral region
- C. Erosions of the dental enamel
- D. Pallor of the conjunctival mucosa
- E. Holosystolic murmur at the left lower sternal border

True: A **Predicted:** A ✓

Problem 9: Reasoning (raw_model (0)) I

The patient is a 23-year-old woman who is embarrassed about the appearance of her nails. The key detail here is that the physician has been shown a photograph of the nails. This suggests that the condition is related to nail appearance rather than systemic symptoms or other physical findings.

Option A: Silvery plaques on extensor surfaces - This could refer to the characteristic appearance of psoriasis, which often presents as silvery scales on the skin, including the extensor surfaces of the limbs. Psoriasis can also affect the nails, causing changes such as pitting, discoloration, or lifting of the nail plate from the nail bed. This is a plausible association with nail appearance.

Option B: Flesh-colored papules in the lumbosacral region - This finding is more likely to be associated with conditions like molluscum contagiosum or other dermatological conditions affecting the skin, not specifically nail changes.

Problem 9: Reasoning (raw_model (0)) II

Option C: Erosions of the dental enamel - This would be more indicative of a systemic condition affecting the teeth and would not typically be associated with nail changes.

Option D: Pallor of the conjunctival mucosa - This finding is more indicative of anemia or other systemic conditions affecting blood perfusion, not specifically nail changes.

Option E: Holosystolic murmur at the left lower sternal border - This finding is more indicative of a cardiac condition and would not typically be associated with nail changes.

Given the context of nail appearance and the options provided, the most likely additional finding related to the patient's nail condition would be silvery plaques on extensor surfaces, which is characteristic of psoriasis.

Therefore, the most likely additional finding is:

Problem 9: Observation Coverage (raw_model (0)) I

Coverage score: $2/10 = 20.0\%$

Overall analysis:

The chosen hypothesis (A: silvery plaques on extensor surfaces / psoriasis) accounts for the explicit link between nail-related findings and psoriasis features, including nails changes. Several raw details about the patient and the other option descriptions are not explicitly connected to this hypothesis within the trace, so they remain unaddressed. The trace does provide a direct, explicit connection for the key detail about psoriasis features and the final conclusion, but it does not logically tie the other enumerated details to psoriasis; thus, some supporting observation details are not explained by the hypothesis. **Observation details:**

Problem 9: Observation Coverage (raw_model (0)) II

OK?	Detail	Evidence
✗	The patient is a 23-year-old woman	
✗	Embarrassed about the appearance of her nails	
✗	The physician has been shown a photograph of the nails	
✗	This suggests that the condition is related to nail appearance rather than systemic symptoms or other physical findings	
✓	Silvery plaques on extensor surfaces	<i>"This is a plausible association with nail appearance."</i>
✗	Flesh-colored papules in the lumbosacral region	

Problem 9: Reasoning (checkpoint-2560) I

The patient's complaint is about the appearance of her nails, which suggests a dermatological issue. The options provided include various skin conditions and systemic manifestations.

Option A: Silvery plaques on extensor surfaces could refer to psoriasis, which often presents as silvery scales on the elbows, knees, and other extensor surfaces. This is a common dermatological condition that can affect nail appearance.

Option B: Flesh-colored papules in the lumbosacral region typically refer to seborrheic keratoses or other benign skin growths, which are not directly related to nail changes.

Option C: Erosions of the dental enamel would be more indicative of a systemic condition affecting oral health rather than nail appearance.

Option D: Pallor of the conjunctival mucosa might indicate anemia but does not directly relate to nail appearance.

Problem 9: Reasoning (checkpoint-2560) II

Option E: A holosystolic murmur at the left lower sternal border would suggest a cardiac issue, which is unrelated to nail appearance.

Given the context of the patient's complaint about nail appearance, the most likely additional finding would be silvery plaques on extensor surfaces, which is characteristic of psoriasis.

Problem 9: Observation Coverage (checkpoint-2560) I

Coverage score: $8/9 = 88.9\%$

Overall analysis:

The chosen hypothesis (psoriasis with silvery plaques on extensor surfaces) adequately accounts for the key observation that nail appearance prompted a dermatological interpretation and is directly linked to the most plausible associated finding described in the trace. Several peripheral details (B–E) are acknowledged as not directly related to nail changes, while the primary link between nail appearance and psoriasis is explicitly stated. The trace connects the core observation to the hypothesis, but some minor contextual options are not used to justify the hypothesis, which is appropriate. Overall, the hypothesis is well-supported by the central details but excludes unrelated data with no explanation required.

Observation details:

Problem 9: Observation Coverage (checkpoint-2560) II

OK?	Detail	Evidence
✓	The patient's complaint is about the appearance of her nails.	<i>"The patient's complaint is about the appearance of her nails, which suggests a dermatological issue."</i>
✓	This complaint suggests a dermatological issue.	<i>"which suggests a dermatological issue."</i>
✗	Options include various skin conditions and systemic manifestations.	
✓	Option A describes silvery plaques on extensor surfaces, which could refer to psoriasis and can affect nail appearance.	<i>"Option A: Silvery plaques on extensor surfaces could refer to psoriasis, which often presents as silvery scales on the elbows, knees, and other extensor surfaces. This is a common dermatological condition that can</i>